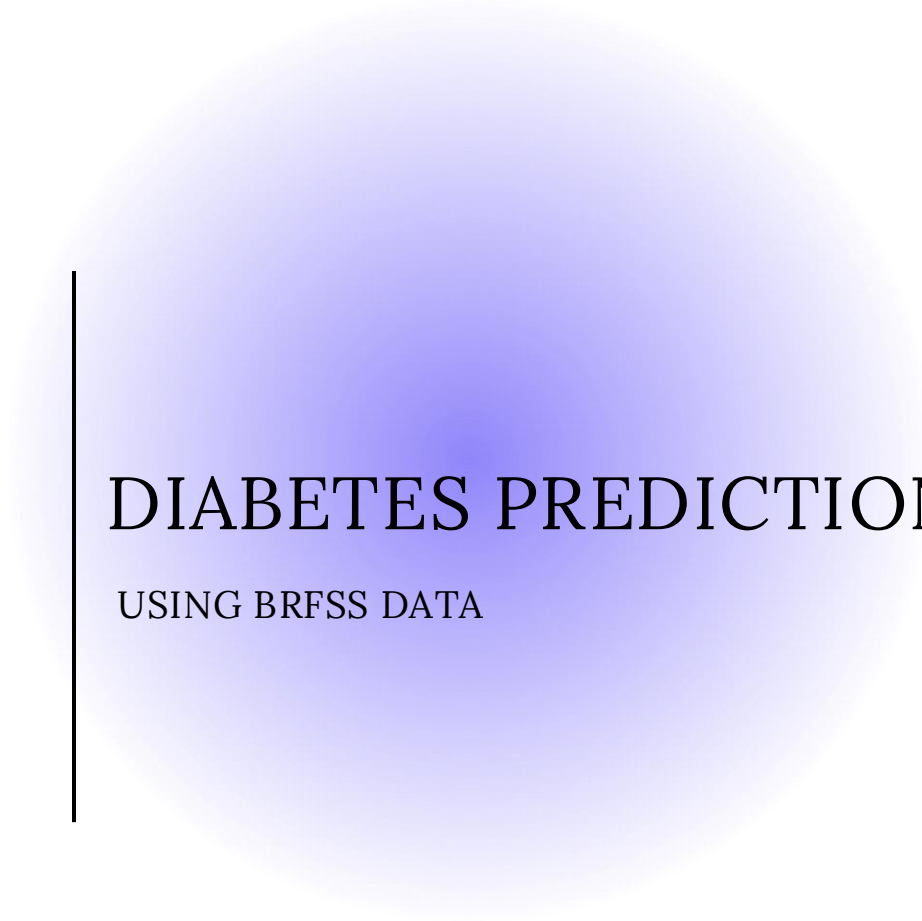




DIABETES PREDICTION USING MACHINE LEARNING

USING BRFSS DATA

Lulu Rashid Massasi (lr-2026) ,Akhil Arakkal, Diego Quispe

PROBLEM & STAKEHOLDER

- **Problem:**

Diabetes is widespread, but early detection is difficult at a population level. Many people at risk are not identified until symptoms appear, and clinical screening requires time, cost, and access to healthcare. There is a need for a scalable way to flag high-risk individuals using readily available data like surveys.

- **Stakeholder:**

- Public health researchers and Centers for Disease Control and Prevention (CDC) data users
- Policymakers and healthcare providers responsible for prevention programs

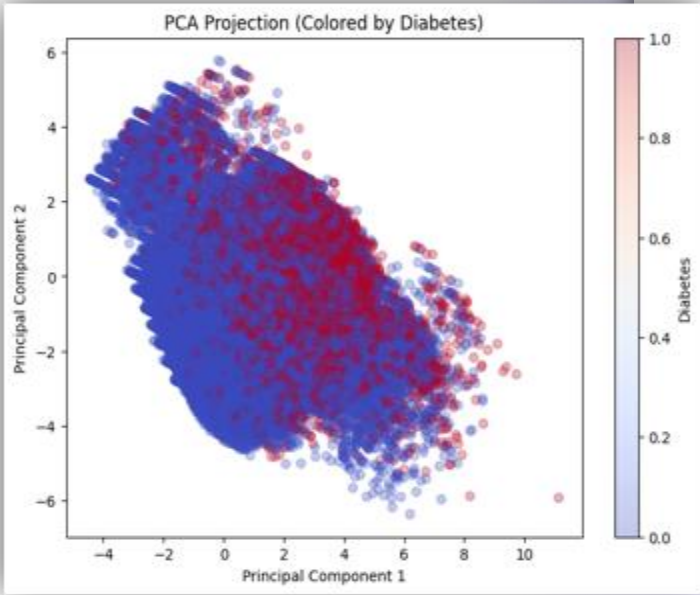
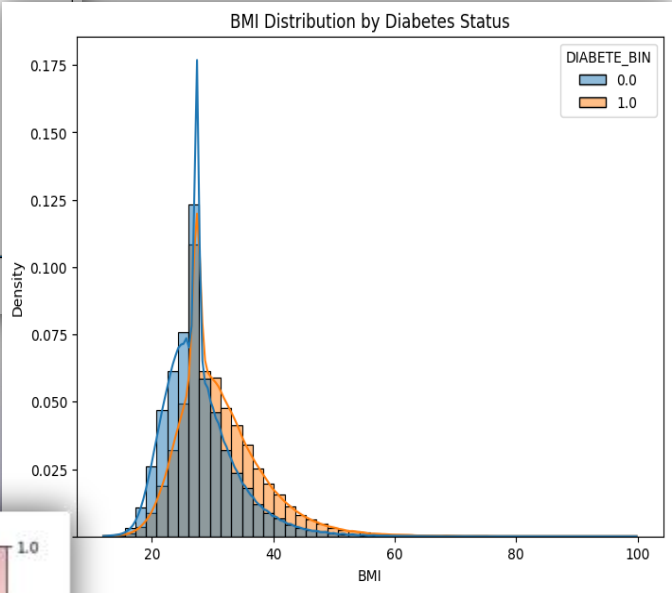
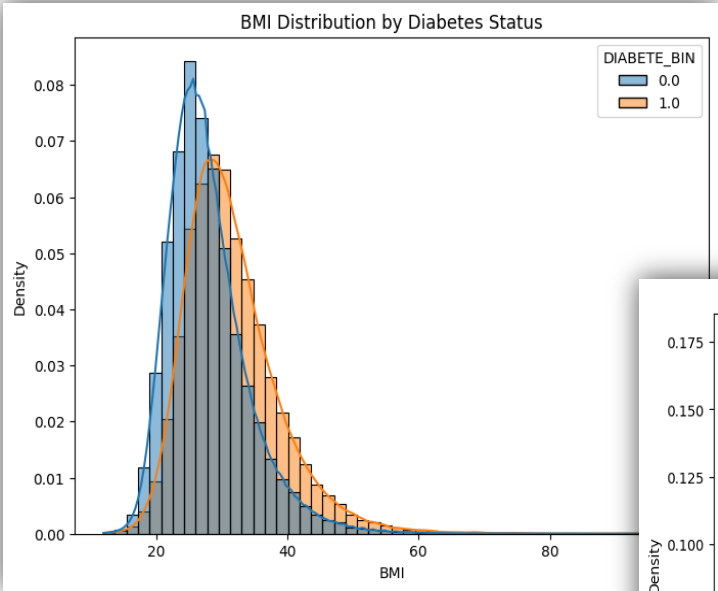
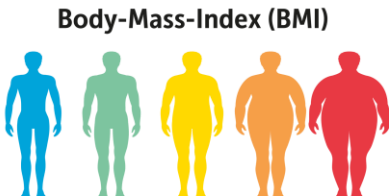
ENVISIONED SOLUTION

- Use BRFSS 2024 survey data to predict Diabetes.
 - Identifying key risk factors influencing Diabetes.
 - Build machine learning models (random forest, XGBoost, DNN)
 - Provide insights to support public health decision-making
- 

DATA

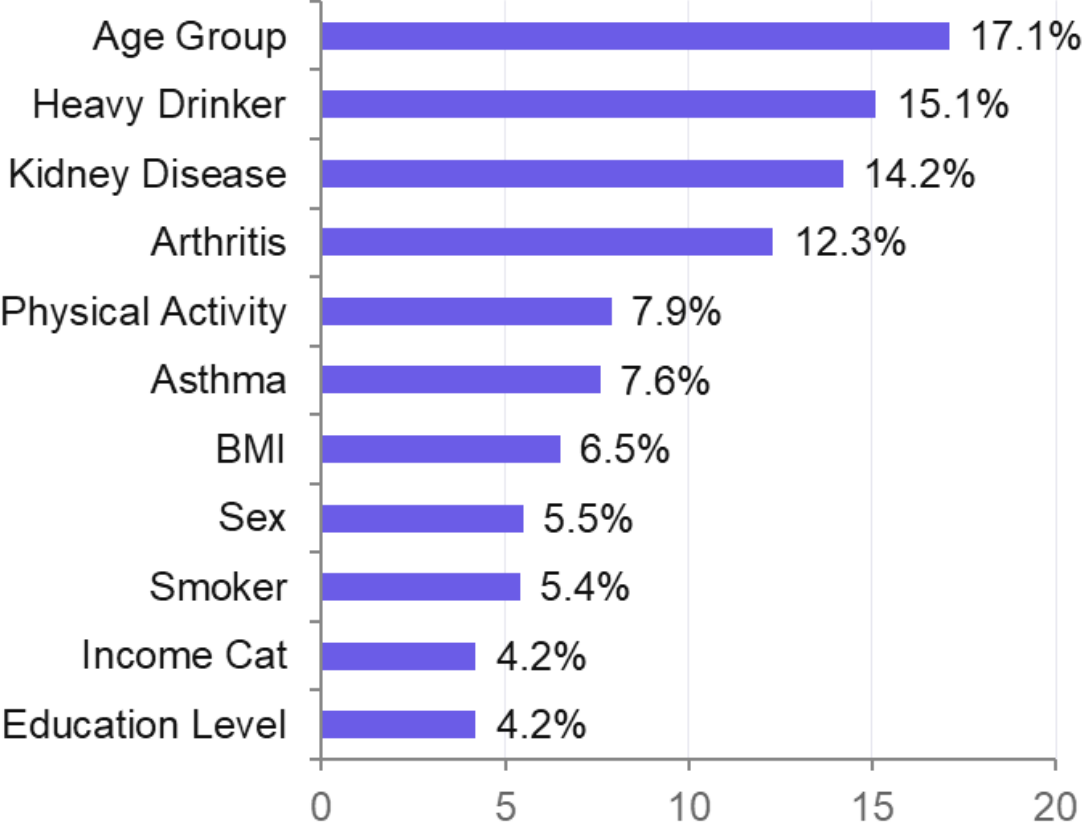
- Source: CDC - BRFSS - Survey Data & Documentation
- Number of observations: ~ 1.3 million (~457,669 - 2024)
- Proportion: 84% (No) - 16%(Yes)

Group	Variables
Demographics	Age sex education
Health Behaviors	Smoking physical_activity Drinking
Health Conditions	asthma kidney arthritis
Physical / Clinical Measure	bmi



BASELINE VS TUNED XGBOOST

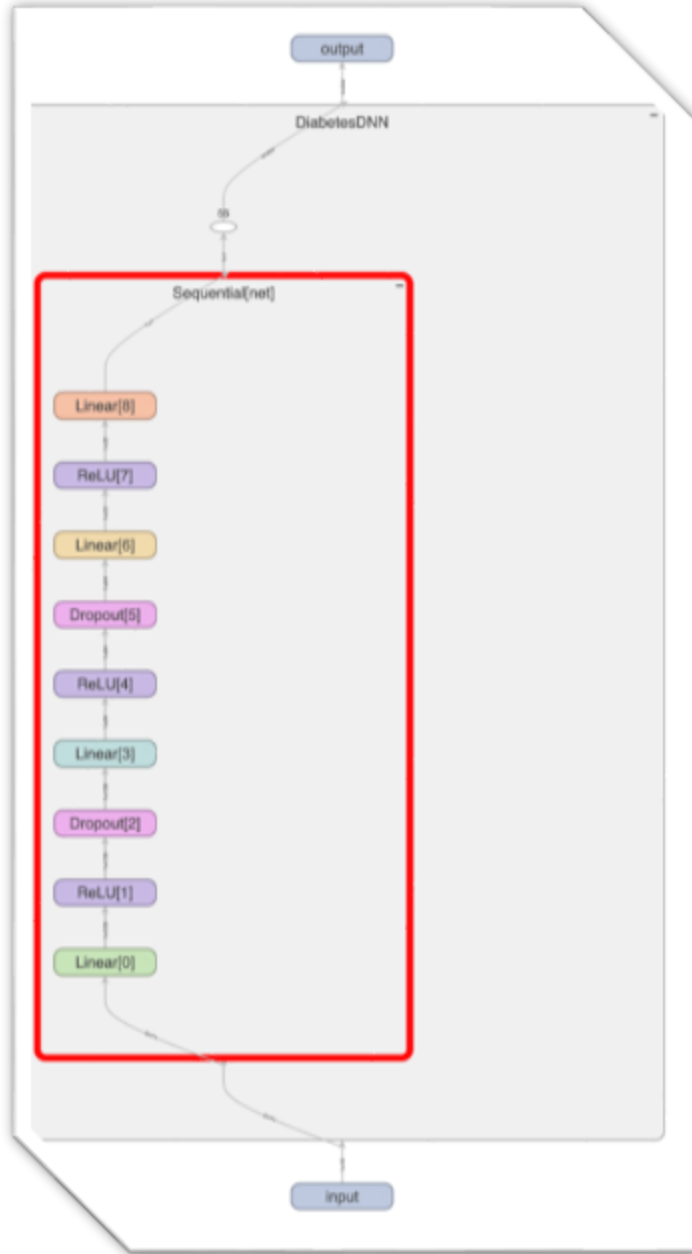
Top Feature Importance (XGBoost Tuned)



MODELS		
Macro	Recall	XgBoost
0.50	0.50	
ROC-AUC:0.5011		
Macro	Recall	RF
0.58	0.61	
ROC-AUC: 0.69		

TUNED MODEL		
Macro	Recall	XgBoost
0.60	0.66	
ROC –AUC: 0.73		

MODELS (2022-2024)



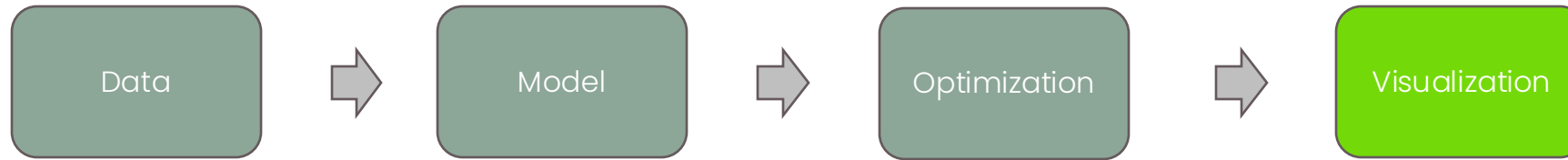
Model	Macro F1	Recall	Best Strength
Logistic Regression	0.60	0.72	Strong baseline stability
Random Forest	0.60	0.70	Good class separation
XGBoost	0.60	0.76	<i>Best overall balance</i>
XGBoost + SMOTE	0.61	0.72	Better minority balance
DNN (no year)	0.434	0.764	<i>High recall sensitivity</i>
DNN + Optuna (threshold tuned)	0.448	0.610	Improved F1 with calibrated threshold
DNN + Year feature	0.434	0.767	<i>Slight recall gain</i>

CHALLENGES

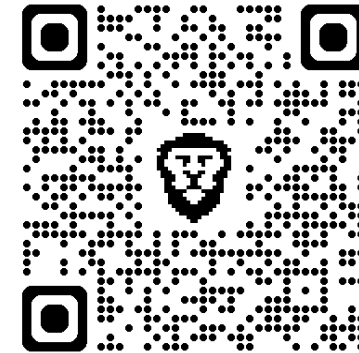
1. **Class imbalance:** Diabetes cases were underrepresented (*used SMOTE*)
2. **Compute cost:** RandomizedSearchCV (*used different models and architectures and OPTUNA*)
3. **Missing data:** Imputation mean (**different imputation could be use**)

WHERE DO WE GO FROM HERE?

1.



<https://riskapp2.streamlit.app/>



2. New Architectures